MA3236 NONLINEAR PROGRAMMING
Semester 1, AY 2026/2027
Tutorial 9

Review: Lagrangian duality (L18)

Dual problem. All problem functions are finite on $\mathbf{R}^n$, and all constraints are explicit:

$$p^* = \inf\{f(\mathbf{x}) : \mathbf{g}(\mathbf{x}) = \mathbf{0},\ \mathbf{h}(\mathbf{x}) \le \mathbf{0}\}, \qquad W = \mathbf{R}^m \times \mathbf{R}^p_+.$$

The equality multipliers are free; inequality multipliers are nonnegative. Set

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{x} \in \mathbf{R}^n} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}) + \boldsymbol{\nu}^T \mathbf{h}(\mathbf{x}), \qquad d^* = \sup_{(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in W} \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}).$$

For fixed multipliers, take the *global* inner infimum over $\mathbf{R}^n$.

Weak duality and objective certificate. For primal feasible $\mathbf{x}$ and $\mathbf{w} = (\boldsymbol{\lambda}, \boldsymbol{\nu}) \in W$,

$$\theta(\mathbf{w}) \le L(\mathbf{x}, \mathbf{w}) \le f(\mathbf{x}) \quad \Longrightarrow \quad \theta(\mathbf{w}) \le d^* \le p^* \le f(\mathbf{x}).$$

- No convexity or constraint qualification is needed. When the values are finite, distinguish the optimal gap $p^* - d^*$ from the computable gap $G = f(\mathbf{x}) - \theta(\mathbf{w})$:

  $$0 \le \max\{p^* - d^*,\ f(\mathbf{x}) - p^*,\ d^* - \theta(\mathbf{w})\} \le G.$$

  Thus $G = 0$ certifies both global optima; $G \le \varepsilon_{\text{out}}$ certifies objective accuracy.

- If $p^* = -\infty$, then $\theta = -\infty$ throughout W. If $d^* = +\infty$, the primal problem is infeasible. The converses need not hold.

Effective domain, finiteness, attainment, uniqueness.

$$\operatorname{dom} \theta = \{\mathbf{w} : \theta(\mathbf{w}) > -\infty\}, \qquad \mathcal{F}_D = W \cap \operatorname{dom} \theta.$$

The pointwise infimum of affine functions is concave, so θ is concave even for a nonconvex primal problem. When the inner problem is separable, minimize its terms individually. In particular,

$$\inf_{z \in \mathbf{R}} (az^2 + bz + c) = \begin{cases} c - b^2/(4a), & a > 0, \\ c, & a = b = 0, \\ -\infty, & a < 0 \text{ or } (a = 0,\ b \ne 0). \end{cases}$$

A finite infimum may be unattained or have many minimizers. Likewise, the dual supremum may be unattained or have many maximizers. The dual function may be nonsmooth.

Review: strong duality and saddle points (L19)

Strong duality. Strong duality means $d^* = p^* \in \mathbf{R}$. Since the multiplier supremum enforces the constraints, this is an exchange of optimization order:

$$p^* = \inf_{\mathbf{x}\in\mathbf{R}^n} \sup_{\mathbf{w}\in W} L(\mathbf{x},\mathbf{w}) \;=\; \sup_{\mathbf{w}\in W} \inf_{\mathbf{x}\in\mathbf{R}^n} L(\mathbf{x},\mathbf{w}) = d^*.$$

For convex f, h_j and affine $\mathbf{g}(\mathbf{x}) = \mathbf{B}\mathbf{x}-\mathbf{b}$, with $p^* \in \mathbf{R}$, the convex strong-duality theorem gives

$$\underbrace{\exists\, \widehat{\mathbf{x}}:\ \mathbf{B}\widehat{\mathbf{x}} = \mathbf{b},\quad h_j(\widehat{\mathbf{x}}) < 0\ (\forall j)}_{\text{Slater's condition}} \quad\Longrightarrow\quad d^* = p^* \text{ and a dual optimizer exists.}$$

- Convexity and Slater are sufficient, not necessary. Convexity or Slater alone does not ensure zero gap.
- Primal attainment is a separate issue. For example, $\inf_{x\geq 0} e^{-x} = 0$ has no minimizer, although convexity and Slater hold.
- For linear programming, strict feasibility is unnecessary:

  $$\min\{\mathbf{c}^T\mathbf{x} : \mathbf{B}\mathbf{x} = \mathbf{b},\ \mathbf{x}\geq\mathbf{0}\} \quad\leftrightarrow\quad \max\{-\mathbf{b}^T\boldsymbol{\lambda} : \mathbf{c}+\mathbf{B}^T\boldsymbol{\lambda}\geq\mathbf{0}\}.$$

  If either side is feasible with finite optimal value, both optima are attained and equal. An unbounded side forces the other side to be infeasible.

Global saddle point. A global saddle point $(\mathbf{x}^*, \mathbf{w}^*)$, with $\mathbf{w}^* \in W$, satisfies

$$L(\mathbf{x}^*,\mathbf{w}) \leq L(\mathbf{x}^*,\mathbf{w}^*) \leq L(\mathbf{x},\mathbf{w}^*) \qquad \forall\ \mathbf{x}\in\mathbf{R}^n,\quad \mathbf{w}\in W.$$

These inequalities are equivalent to primal and dual optimality of the pair with $f(\mathbf{x}^*) = \theta(\mathbf{w}^*)$. This equivalence needs no convexity. For differentiable data, write KKT as

$$\nabla_{\mathbf{x}} L = \mathbf{0},\quad \mathbf{g} = \mathbf{0},\quad \mathbf{h}\leq\mathbf{0},\quad \boldsymbol{\nu}\geq\mathbf{0},\quad \nu_j h_j = 0\ (\forall j).$$

- Attained primal and dual optima with zero gap imply KKT. Conversely, KKT implies a global saddle point when f, h_j are convex and $\mathbf{g}$ is affine, because stationarity then globally minimizes $L(\cdot, \mathbf{w}^*)$.
- In a nonconvex problem, KKT alone does not certify zero gap. Even a zero gap alone supplies no saddle point unless both optima are attained.

Primal recovery. Even at a dual optimizer, an arbitrary inner minimizer may be infeasible. Choose $\mathbf{x}^* \in \arg\min_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ and verify

$$\mathbf{g}(\mathbf{x}^*) = \mathbf{0},\quad \mathbf{h}(\mathbf{x}^*)\leq\mathbf{0},\quad (\boldsymbol{\nu}^*)^T\mathbf{h}(\mathbf{x}^*) = 0 \quad\Longrightarrow\quad f(\mathbf{x}^*) = \theta(\boldsymbol{\lambda}^*,\boldsymbol{\nu}^*).$$

1. **A quadratic dual with two kinds of multipliers.** Consider

$$\begin{aligned} \min_{\mathbf{x}\in\mathbf{R}^2} \quad & f(\mathbf{x}) = 2x_1^2 + x_2^2 - 4x_1 - 6x_2 \\ \text{s.t.} \quad & g(\mathbf{x}) = x_1 + x_2 - 4 = 0, \\ & h(\mathbf{x}) = -x_1 + 2x_2 - 4 \le 0. \end{aligned}$$

   (a) Write $L(\mathbf{x}, \lambda, \nu)$, state the dual multiplier restrictions, and derive $\theta(\lambda, \nu)$.

   (b) Solve the dual problem and recover a primal minimizer. Check if strong duality holds by exhibiting a Slater point.

   (c) Verify the saddle inequalities directly at this primal–dual pair.

2. **Constraint formulation, KKT, and the duality gap.** Let $f(x) = x^4 - 2x^2$ for $x \in \mathbf{R}$.

   (a) Consider $\min_x f(x)$ subject to $x^2 \le 0$. Find p^*, derive $\theta(\nu)$ for every $\nu \ge 0$, and determine d^* and all optimal multipliers. Check whether the convex strong-duality theorem applies.

   (b) Replace the inequality by $x = 0$. Find the dual optimal value and all optimal multipliers, and compare the two duality gaps. Determine whether $(x, \lambda) = (0, 0)$ satisfies KKT and whether a global Lagrangian saddle point exists for this equality formulation.

3. **A dual with a restricted effective domain.** Consider

$$\min_{\mathbf{x}\in\mathbf{R}^3} f(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2 \quad \text{s.t.} \quad g(\mathbf{x}) = 2x_1 + x_2^2 + x_3 - 4 = 0.$$

   (a) Derive $\theta(\lambda)$, including every λ for which $\theta(\lambda) = -\infty$.

   (b) Solve the dual problem and determine its Lagrangian-subproblem minimizer set at λ^*.

   (c) Recover all primal global minimizers and certify zero gap. Explain why the usual convex Slater theorem is not the justification.